

MITRE ATT&CK; Research Internship Report

Group Leader Selection Program

Intern ID: 2164

Assigned Campaign: LockBit Group Campaign

Assigned Software Range: S0640 – S0650

This report is prepared using MITRE ATT&CK; official references and open-source cyber threat intelligence research. The objective of this report is to analyze the LockBit ransomware ecosystem and research software entries assigned within the S0640–S0650 range.

1. Campaign Research – LockBit Group

Campaign Overview:

LockBit is one of the most active ransomware-as-a-service (RaaS) groups observed globally. The group targets enterprises, government institutions, healthcare organizations, financial companies, manufacturing industries, and critical infrastructure. LockBit operators use affiliate-based attacks where affiliates deploy ransomware while the core developers maintain the malware ecosystem.

Threat Actor Association:

LockBit is associated with financially motivated cybercriminal groups operating globally. The ransomware family evolved through multiple versions, including LockBit 2.0 and LockBit 3.0 (LockBit Black).

Timeline:

- 2019 – Initial LockBit ransomware observed.
- 2021 – LockBit 2.0 introduced automated propagation and advanced evasion.
- 2022 – LockBit 3.0 launched bug bounty style affiliate program.
- 2024–2025 – International law enforcement operations disrupted parts of LockBit infrastructure.

Objectives of Attack:

- Financial extortion through ransomware encryption.
- Data theft and double extortion.
- Operational disruption of organizations.

Targeted Sectors and Countries:

LockBit campaigns impacted victims across the United States, United Kingdom, India, Europe, Latin America, and Asia-Pacific regions. Targeted sectors include healthcare, education, finance, manufacturing, logistics, and energy.

Attack Flow:

1. Initial access through phishing, exposed RDP, or stolen credentials.
2. Privilege escalation and lateral movement inside the network.

3. Credential dumping and system discovery.
4. Data exfiltration to attacker-controlled infrastructure.
5. File encryption and ransom demand deployment.

MITRE ATT&CK; Techniques Commonly Used:

- T1566 – Phishing
- T1021 – Remote Services
- T1059 – Command and Scripting Interpreter
- T1003 – OS Credential Dumping
- T1486 – Data Encrypted for Impact
- T1041 – Exfiltration Over C2 Channel
- T1490 – Inhibit System Recovery

Real-World Incidents:

LockBit operations affected hospitals, educational institutions, and government agencies. Several incidents involved large-scale data leaks and ransomware deployment against enterprise infrastructure.

Detection Opportunities:

- Monitor abnormal PowerShell and command shell activity.
- Detect suspicious RDP logins and lateral movement behavior.
- Monitor mass file modification and encryption patterns.
- Detect credential dumping attempts and privilege escalation behavior.

Defensive Recommendations:

- Enable multi-factor authentication (MFA).
- Regularly patch internet-facing systems.
- Restrict RDP access and apply network segmentation.
- Maintain offline backups and endpoint detection solutions.
- Conduct continuous threat hunting and log monitoring.

2. Software Research (S0640 – S0650)

S0640 – Avaddon

Type: Ransomware

Supported Platforms: Windows

Description: Encrypts files and disables recovery mechanisms.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1486, T1490, T1547

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0641 – Kobalos

Type: Backdoor

Supported Platforms: Linux

Description: Linux backdoor used against high-profile systems.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1059, T1573, T1090

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0642 – Cuba

Type: Ransomware

Supported Platforms: Windows

Description: Encrypts enterprise systems for financial extortion.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1486, T1059

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0643 – Doki

Type: Backdoor

Supported Platforms: Linux

Description: Cloud-based malware using dynamic DNS techniques.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1105, T1071

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0644 – EnvyScout

Type: Dropper

Supported Platforms: Windows

Description: Used for malware delivery via phishing campaigns.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1204, T1105

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0645 – FlashFlood

Type: Botnet Malware

Supported Platforms: Linux

Description: IoT malware targeting Linux devices.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1498, T1071

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0646 – GOODLUCK

Type: Ransomware

Supported Platforms: Windows

Description: Encrypts files and deletes shadow copies.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1486, T1490

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0647 – HEAVYHAND

Type: Downloader

Supported Platforms: Windows

Description: Used for downloading secondary payloads.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1105, T1027

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0648 – Kwampirs

Type: Trojan

Supported Platforms: Windows

Description: Healthcare-focused malware for espionage.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1055, T1082

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0649 – MosaicLoader

Type: Loader

Supported Platforms: Windows

Description: Malware loader distributed through pirated software.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1105, T1204

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

S0650 – QakBot

Type: Banking Trojan

Supported Platforms: Windows

Description: Advanced malware supporting credential theft and lateral movement.

Threat Actor Association: Linked with financially motivated cybercriminal groups and ransomware operations.

ATT&CK; Techniques: T1053, T1003, T1021

Execution Method: Phishing attachments, malicious downloads, exploit-based execution, or scripts.

Persistence: Registry keys, scheduled tasks, or startup modifications.

Privilege Escalation: Uses system vulnerabilities or token manipulation in some campaigns.

Defense Evasion: Obfuscation, encrypted payloads, anti-analysis techniques, and disabling security tools.

Credential Access: Credential dumping, keylogging, or browser credential theft observed in related campaigns.

Discovery Techniques: Process discovery, network discovery, and file enumeration.

Lateral Movement: SMB, RDP, or remote services used for movement across networks.

Command & Control: Uses encrypted communication channels and remote servers.

Exfiltration: Sensitive data transferred to attacker infrastructure before encryption or impact operations.

Impact Analysis: Data loss, operational disruption, financial extortion, and reputational damage.

Indicators of Compromise: Suspicious registry changes, unknown scheduled tasks, unusual outbound traffic, encrypted files, and malicious hashes.

Detection & Mitigation: EDR monitoring, network segmentation, MFA, patch management, backup strategy, and SIEM-based threat detection.

3. References & Sources

- MITRE ATT&CK; Official Website – <https://attack.mitre.org/>
- MITRE ATT&CK; Software Database – <https://attack.mitre.org/software/>
- MITRE ATT&CK; Campaign Database – <https://attack.mitre.org/campaigns/>
- MITRE ATT&CK; Technique Matrix – <https://attack.mitre.org/techniques/enterprise/>
- Open-source cybersecurity threat intelligence reports and malware research articles.

Conclusion:

This research demonstrates analysis of ransomware operations, malware behavior, ATT&CK; mappings, and adversary techniques associated with enterprise cyber threats. The report focuses on threat intelligence, technical understanding, and defensive cybersecurity practices aligned with MITRE ATT&CK; methodology.